WNDRVR
SUPPORT NETWORK

VXWORKS 7 TFTP PROGRAMMER'S GUIDE, SR0650

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VxWorks 7 TFTP Programmer's Guide, SR0650

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1. OVERVIEW

1. TFTP for VxWorks 7

The VxWorks® 7 TFTP Programmer's Guide describes how the VxWorks Trivial File Transfer Protocol (TFTP) reads and writes files on a remote server.

TFTP is a basic form of FTP; it uses a minimal amount of memory, cannot list directory contents, and has no authentication or encryption mechanisms. Unlike FTP, which uses the TCP transport layer protocol (port 21), TFTP uses the UDP (port 69) as its transport protocol.

Wind River TFTP implements the following features:

- a TFTP server and client conforming to RFC 1350
- a TFTP server that listens to both IPv4 and IPv6 connections
- multiple client connections
- both octet and netascii modes
- an API to enable TFTP client operations
- communication over IPv4 or IPv6
- a shell command to transfer files to and from the target

Source Code Files

The source code files for TFTP are in the following directory:

installDir **/vxworks-7/pkgs_v2/net/ipnet-***x.x.x.x* **/app/tftp-***x.x.x.x.* **/**

2. Supported RFCs for TFTP

Trivial File Transfer Protocol (TFTP) request for comments (RFCs) contain standards and design information defined by the Internet Engineering Task Force (IETF).

TFTP for VxWorks 7 complies with the following IETF RFC:

- RFC 1350: *The TFTP Protocol (Revision 2)*

2. TFTP CONFIGURATION AND BUILD

1. TFT Configuration and Build Workflow

Your Wind River VxWorks 7 installation includes source code but does not include pre-built kernel libraries.

Before beginning any VxWorks development, you must build the VxWorks kernel libraries (VSB project). You must also build the VxWorks run-time kernel image based on the kernel libraries (VIP).

For specific instructions on how to configure and build a VSB and a VIP, and how to create a downloadable kernel module (DKM) or a real-time process (RTP) application, see the VxWorks 7 Configuration and Build Guide.

For a list of available source configuration options for TFTP, see TFTP Source Build Configuration Options (VSB) on page 2.

For a list of available kernel configuration components and parameters for TFTP, see Kernel Configuration Components (VIP) on page 2.

2. TFTP Source Build Configuration Options (VSB)

Trivial File Transfer Protocol (TFTP) configuration includes VxWorks source build (VSB) options.

To include TFTP support in your VSB, include the following option:

Table 1. TFTP VSB Options

Component	Description
IPNET_TFTP	Enable TFTP in the VSB.

3. TFTP Kernel Configuration Components (VIP)

Trivial File Transfer Protocol (TFTP) includes VxWorks image project (VIP) kernel configuration components.

Table 1. TFTP Components

Component	Description
INCLUDE_IPTFTPC	Include the TFTP client in your VIP. The TFTP client works over both IPv4 and IPv6.
INCLUDE_IPTFTPS	Include the TFTP server in your VIP. The TFTP server works for IPv4 and IPv6. To enable the TFTP upload function (disabled by default), see TFTP Server Upload Function on page 12.

Component	Description
INCLUDE_IPTFTP_COMMON	Include the TFTP server common configuration parameters. See TFTP Configuration Parameters on page 3.

4. TFTP Optional Kernel Configuration Components

You can modify the Trivial File Transfer Protocol (TFTP) client behavior with a number of optional shell command components.

Table 1. TFTP Optional Components

Component	Description
INCLUDE_TFTP_CLIENT	Include the TFTP API component for backward compatibility with a previous release of the Wind River Network Stack, and to carry forward certain routines not provided in the TFTP client. For more information on this component, see the reference entry for **tftpLib**.
INCLUDE_BOOT_TFTP_LOADER	Include the boot application to load files with TFTP at startup. Support routines for the TFTP bootrom loader are in *installDir* **/vxworks-7/pkgs_v2/os/utils/bootapp-***x.x.x.x* **/src/bootTftpLoad.c.**
INCLUDE_IPTFTP_CLIENT_CMD	Include the TFTP client shell command (tftp).
INCLUDE_IPD_CMD	Include the daemon control shell command (ipd).
INCLUDE_IPCOM_SYSLOGD_CMD	Include the system logging shell command (syslog).
INCLUDE_IPCOM_SYSVAR_CMD	Include the system variables shell command (sysvar)

5. TFTP Configuration Parameters

You can modify the Trivial File Transfer Protocol (TFTP) client behavior with a number of configuration parameters and system variables.

Table 1. TFTP Configuration Parameters

Parameter Name, System Variable Name, and Description	Default and Type
TFTP_CFGFILE The full path of the configuration file that VxWorks will use to configure the TFTP parameters. When no file is specified (default setting), the TFTP parameter values are based on the VIP configuration. For more information, see About TFTP File-Based Configuration on page 6.	"" char *
TFTPS_DELAYEDSTART When set to FALSE, the TFTP server is automatically started upon booting VxWorks. When set to TRUE, the TFTP server is not started upon booting VxWorks. The TFTP server can be started with an ipd command. For more information, see About TFTP File-Based Configuration on page 6.	FALSE BOOL
TFTPS_RETRIES iptftp.retries The number of times TFTP will resend a message if it does not receive an acknowledgment from the peer. Both the client and the server use this parameter.	"2" char *
TFTPS_RETRANSMIT_TIMEOUT iptftp.timeout The time to wait before the first retry is sent, in seconds. Further retries use an exponential back-off algorithm. This parameter is used by both the client and server.	"5" char *
TFTPS_DIRS iptftp.dir The TFTP server working directory.	IPCOM_FILE_ROOT "tftpdir" char *
iptftp.srvPort TFTP server listening port.	"69" char *
iptftp.clntSrvPort TFTP port used by the client to send packets.	"69" char *

Parameter Name, System Variable Name, and Description	Default and Type
iptftps.allow.upload Specifies if TFTP write requests are permitted. Write requests are disabled by default. To enable the TFTP upload function, set to enabled. For more information, see TFTP Server Upload Function on page 12.	"" char *
TFTPS_TASK_PRIORITY This parameter sets the task priority of the Trivial File Transport Protocol (TFTP) server network service task. Changing the value can have risks.	"50" char *
TFTPS_DEBUG_ENABLE To debug the TFTP server network service task, set this value to "1".	"0" char *

3. FILE-BASED CONFIGURATION

1. About TFTP File-Based Configuration

By default, Trivial File Transfer Protocol (TFTP) parameter values are set at VIP configuration time. The file-based configuration functionality allows you to reconfigure TFTP parameter values at boot time.

You reconfigure TFTP parameter values with a list of system variables stored in a text-based configuration file. The file path and file name are specified by a VIP parameter. This configuration method provides an easy way to adapt a system to new networking environments by simply updating the configuration file contents and then transferring the file to the target before starting TFTP .

A delayed start functionality is also available to prevent TFTP from starting automatically during the VxWorks boot process. You can start TFTP after VxWorks has booted or at runtime from your application.

2. Enabling and Disabling TFTP File-Based Configuration

To enable the configuration of Trivial File Transfer Protocol (TFTP) from a configuration file, you must specify the full path to the configuration file as part of the TFTP_CFGFILE VIP parameter.

About This Task

To disable the configuration of TFTP from a configuration file, set the TFTP_CFGFILE VIP parameter to NULL (**""**). TFTP configuration is then based on the VIP configuration.

For information on configuring the TFTP_CFGFILE VIP parameter, see TFTP Configuration Parameters on page 3.

For information on startup scenarios for TFTP, see TFTP Startup Scenarios on page 9.

For information on operational requirements for the target and TFTP, see Operational Requirements on page 10.

3. Sample TFTP Configuration File

A sample Trivial File Transfer Protocol (TFTP) configuration file that contains the list of valid system variables can be configured from a configuration file.

The sample configuration file is at:

installDir **/vxworks-7/pkgs_v2/net/ipnet-***x.x.x.x* **/app/tftpt-***x.x.x.x* **/sample_cfg/etc/tftp.conf**

> Note: Only the specified system variables in the sample file are configurable from a configuration file.
>
> For a list of TFTP system variables not configurable from the configuration file, see TFTP Parameters Not Configurable from a Configuration File on page 7.

To correlate TFTP VIP parameter names to TFTP system variables listed in the sample configuration file, see TFTP Configuration Parameters on page 3.

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Note: To prevent unnecessary reconfiguration of the VIP parameters, the configuration file should only contain system variables that must be changed. The remaining parameter values will be taken from the VIP configuration.

4. TFTP Parameters Not Configurable from a Configuration File

The **iptftps.allow.upload** parameter listed in Trivial File Transfer Protocol (TFTP) Configuration Parameters is not configurable from the configuration file.

You must configure this parameter from a startup script at boot time or programmatically at runtime by creating and changing the value of the system variable. For more information, see TFTP Server Upload Function on page 12.

5. TFTP Configuration File Format

The Trivial File Transfer Protocol (TFTP) configuration file is a text-based file you can edit with a standard text editor.

To configure a parameter from a configuration file, a valid system variable name must be followed by a valid value (system variable / value pair) and be on a single line.

A configuration file can also contain comments.

Note: The configuration file does not accept Escape characters (\).

System Variable and Value Pair

A system variable/value pair consists of a system variable name and a value. Valid system variables are provided in the sample configuration file.

A system variable/value pair can start in any column of a given line. A system variable and name must be separated by least one space or tab.

Each line can only contain a single system variable/value pair.

Boolean, integer, floating point, and string values can be listed with or without double quotation marks. Example valid line content follows:

```
systemVariableName_A  TRUE
systemVariableName_B  2
systemVariableName_C  192.168.1.1
systemVariableName_D  String
```

```
systemVariableName_A  "TRUE"
systemVariableName_B  "2"
systemVariableName_C  "192.168.1.1"
systemVariableName_D  "String"
```

Comment Delimiter

The comment delimiter for the configuration file is the pound (**#**) character. For example:

```
# This is a comment line.
```

You can add a comment on the same line as a system variable/name pair. For example:

```
systemVariableName_A  FALSE         # This is a boolean value.
systemVariableName_B  22            # This is an integer value.
systemVariableName_C  192.168.2.2   # This is a floating point value.
systemVariableName_D  String        # This is a string value.
```

6. TFTP Configuration File Processing

All empty lines and comment lines in the Trivial File Transfer Protocol (TFTP) configuration processing file are not processed.

If a configuration file is not found, no processing is done and the system variables retain their values from the VIP configuration.

If a configuration file is located, the system processes the configuration file line-by-line. The values in the configuration file overwrite the values set at VIP configuration time.

Lines with errors such as misspelled system variable names and missing parameter values are ignored (skipped). Hence, a system variable associated with a faulty line entry retains the value set at VIP configuration time.

Note: If a system variable/value pair is listed more than once in a file, the last valid occurrence is used to configure the system variable. No error is reported as part of the diagnostic output.

7. TFTP Diagnostics

You enable and disable diagnostics from within the configuration file. When enabled, the configuration status of the system variables is sent to the console, if a console is available.

To enable diagnostics, set the **cfgfile.diag** variable in the configuration file to **TRUE**. To disable, set to **FALSE**. For example:

```
cfgfile.diags      FALSE
```

When enabled, the diagnostics process generates output and provides status on the action performed on each system variable.

An updated system variable is labelled as **Updated**. The output format is as follows:

`Variable:` *systemVariableName* `Value:` *value* `Result: Updated`

A skipped update for a system variable is labelled as **Unrecognized variable (Skipping)**. The output format is as follows:

`Variable:` *systemVariableName* `Value:` *value* `Result: Unrecognized variable (Skipping)`

8. Verifying TFTP System Variables

To verify the configuration of the Trivial File Transfer Protocol (TFTP) system variables, you can display the system variables and their content from the command shell.

About This Task

To view system variables, use the following command:

```
vxWorks# sysvar list serviceName
```

Note: To access the **sysvar** shell command, you must add the **INCLUDE_IPCOM_SYSVAR_CMD** component to your VIP.

9. TFTP Startup Scenarios

Possible startup scenarios as set by the **TFTP_CFGFILE** and **TFTPS_DELAYEDSTART** VIP configuration parameters.

Table 1. Operational Scenarios of File-Based Configuration for the TFTP Server

TFTP_CFGFILE VIP Parameter Setting	TFTPS_DELAYEDSTART VIP Parameter Setting	Operation
Null file path	FALSE	TFTP starts automatically at boot time. TFTP parameter values are based on the VIP configuration. This is the default configuration.
Valid file path	FALSE	TFTP starts automatically at boot time. TFTP parameter values are based on valid system variable names and values listed in the configuration file.
Null file path	TRUE	TFTP is not started at boot time. TFTP can be started with an ipd start command executed from a startup script or programmatically. TFTP parameter values are based on the VIP configuration.
Valid file path	TRUE	TFTP is not started at boot time. TFTP can be started with an ipd start command executed from a startup script or programmatically. TFTP parameter values are based on valid TFTP system variable names and values listed in the configuration file.

Note: To access the **ipd** shell command, you must add the **INCLUDE_IPD_CMD** component to your VIP.

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10. Operational Requirements

To enable parameter configuration from a file, the target hosting Trivial File Transfer Protocol (TFTP) must include read/write mass storage capability.

A configuration file can be modified by an external host and transferred to the target using one of the following methods:

- By transferring the file from a FTP server to the target.
- By mounting the target's mass storage device as a USB disk and using desktop system facilities to update the configuration file.
- By using a removable mass storage device supported by the target.

4. SHELL COMMANDS

1. About TFTP Shell Commands

The Trivial File Transfer Protocol (TFTP) shell commands include tftp, sysvar, ipd, and syslog.

The shell commands perform the following functions:

You can use the tftp shell command to configure and run the TFTP client.

You can use the sysvar shell command to modify TFTP configuration parameters.

You can use the ipd shell command to start and stop the TFTP server daemon.

You can use the syslog shell command to set the priority for the TFTP syslog debug messages.

To run shell commands, you must access the command-interpreter mode. To access the command-interpreter mode from the kernel shell, enter the following command:

```
-> cmd
[vxWorks *} #
```

To return to the kernel shell, enter the following command:

```
# C
->
```

2. TFTP Client Shell Command

To get help for the tftp command, use the tftp command.

For example:

```
# tftp
usage: tftp [-a] <host [port]> <get|put> <source> [destination]
```

For information on the Trivial File Transfer Protocol (TFTP) client command, see Using the TFTP Client Shell Command on page 15.

3. Setting the TFTP System Variables

You can view all Trivial File Transfer Protocol (TFTP) client sysvar variables from the kernel shell in command-interpreter mode.

To view all TFTP sysvar variables, access the command-interpreter mode from the kernel shell and execute the following command:

```
# sysvar list iptftp
```

To change TFTP parameters with sysvar, use the sysvar set command.

4. TFTP Server Upload Function

The Trivial File Transfer Protocol (TFTP) server upload function is disabled by default.

About This Task

To enable the TFTP upload function, create the **iptftps.allow.upload sysvar** parameter and set the parameter value to **enable** with the following command:

```
# sysvar set -c -o iptftps.allow.upload "enable"
sysvar: iptftps.allow.upload=enable ok
```

To view the current TFTP **sysvar** settings, use the **list** command. For example:

```
# sysvar list iptftp
System variables:
      iptftp.clntSrvPort=69
   iptftp.debug=0
   iptftp.dir=/ram/tftpdir
   iptftp.priority=50
   iptftp.retries=2
   iptftp.srvPort=69
   iptftp.timeout=5
   iptftps.allow.upload=enable
```

5. Stopping and Starting the TFTP Server Daemon

Use the ipd command to stop or start the Trivial File Transfer Protocol (TFTP) server daemon (iptftps).

For example:

```
# ipd list
Services:
...
iptftps      started

# ipd kill iptftps
ipd: kill iptftps ok

# ipd list
Services:
...
iptftps      killed

# ipd start iptftps
ipd: start iptftps ok
```

To view the task information for the TFTP server, run the task command and locate the iptftps task. For example:

```
# task
NAME       ENTRY       TID         PRI    STATUS     PC           SP           ERRNO     DELAY
```

```
...
iptftps   10038560   104e5b70   20    PEND      102f4bbb    10fbfe50    0        0
```

6. Setting the TFTP System Log Priority

Run the syslog list command to get the current priority level for all modules.

```
# syslog list
    syslog facility  priority
    ...
    iptftp Error
```

To list the available syslog priority levels, use the following command:

```
# syslog priority *
syslog: missing <prio>
        priorities: Emerg Crit Error Warning Notice Info Debug Debug2
```

Use the following command syntax to change the priority level for TFTP:

```
# syslog priority iptftp priorityLevel
```

For more information on the syslog shell command, see the VxWorks 7 Network Stack Programmer's Guide.

5. API ROUTINES

1. TFTP Client API Functions

A number of API functions are available to the Trivial File Transfer Protocol (TFTP) client.

For more information, see the reference entries for the TFTP client (iptftp).

The TFTP client public API routines are defined in **iptftpc.c**.

Table 1. TFTP Client API Functions

Fuction	Description
iptftp_client_get( )	Transfer a file to the target from a TFTP server.
iptftp_client_put( )	Transfer a file from the target to a TFTP server.

6. TFTP CLIENT SHELL COMMAND

1. Using the TFTP Client Shell Command

The TFTP client command allows you to transfer files using the TFTP protocol.

The TFTP client command syntax is as follows:

```
tftp [-a] host [port] {get | put} source [destination]
```

The available options for the tftp command are as follows:

-a

Use netascii mode (the default is to use binary/octet mode).

host

The IP address of the host to transfer files with.

port

The port number of the server. If no port is specified, the default is used.

get | **put**

Specify **get** to receive a file or **put** to transmit a file.

source

The name of the file to transmit or receive.

destination

The name of the file after it is transmitted or received.